

Vasundhara D P

✉ dpvasundhara@gmail.com | 📞 +91 79758 57932 | Bengaluru, Karnataka | 🔗 linkedin.com/in/vasundharadp | 🐙 github.com/VasundharaDP123

Summary

Artificial Intelligence and Machine Learning undergraduate with experience building production-ready AI applications using LLMs, RAG, FastAPI, React, and Computer Vision. Skilled in developing scalable backend systems, deploying cloud-based AI solutions, and integrating modern AI technologies into real-world products. Active open-source contributor with freelance experience delivering end-to-end AI solutions.

Education

Sir M. Visvesvaraya Institute of Technology, Bengaluru

Bachelor of Engineering in Artificial Intelligence and Machine Learning

CGPA: 9.1/10 — Class XII: 94.66% — Class X: 98.40%

2023 – Present

Skills

Languages: Python, C, TypeScript, HTML5, CSS3, SQL

Frameworks & Libraries: FastAPI, Flask, React.js, Tailwind CSS, Scikit-Learn, OpenCV, MediaPipe

Databases & Tools: PostgreSQL, SQLite, DBMS, Git, Docker, GitHub Actions, AWS EC2, Linux, Jupyter Notebook, Power BI, DAX

AI/ML: Machine Learning, NLP, Computer Vision, LLMs, LLM APIs, Retrieval-Augmented Generation (RAG)

Backend & Systems: REST APIs, CI/CD

Experience

Freelance AI Developer

2026 – Present

- Developed and deployed **Styvoe**, an AI-powered travel stylist using **FastAPI** and **React** for personalized, weather-aware outfit recommendations.
- Integrated **Open-Meteo API** and optimized **GPT-4o-mini** prompts for culturally intelligent outfit recommendations.
- Integrated the **Amazon Creator API** with backend fallback handling for reliable affiliate product recommendations.
- Implemented real-time streaming using **Server-Sent Events (SSE)** for sub-second interactive responses.

Projects

Netra Rakshaka – Camera-Free Wearable IoT & Ocular Strain Mitigation Platform

- Built a privacy-first **ESP32-S3** smart spectacle using IR, ToF, BMP280 and MPU6050 sensors for fatigue monitoring.
- Developed a real-time **Random Forest** ocular strain classifier with **Groq LLM** clinical advisory and **Socket.IO** telemetry. **Tech Stack:** ESP32-S3, C++, Python, Flask-SocketIO, Scikit-Learn, Groq LLM.

SignWave AI – Real-Time Sign Language Translation Suite

- Built a React, TypeScript, MediaPipe and Flask web app achieving **95%+** hand-tracking accuracy.
- Developed a Custom Gesture Studio, Time-Attack practice game and SVG analytics dashboard.
- Optimized processing to **30 FPS** with sub-**100 ms** latency.

HeatShield – AI-Powered Heatwave Prediction System

- Built an AI-powered heatwave prediction platform using meteorological and AQI data.
- Developed a Random Forest model achieving **92%** prediction accuracy.
- **Tech Stack:** React.js, FastAPI, Python.

DocuMind – Enterprise Retrieval-Augmented Generation

- Built and deployed a RAG application on AWS EC2 with FastAPI.
- Achieved **90%** retrieval accuracy using LLM APIs.
- **Tech Stack:** Python, FastAPI, AWS EC2.

Real-Time Face Mesh Visualization

- Built facial landmark detection using MediaPipe and OpenCV with **98%** accuracy.
- Implemented eye-blink detection and head tracking.
- **Tech Stack:** Python, OpenCV, MediaPipe.

Open Source Contributions

Open Source Contributor – GirlScript Summer of Code (GSSoC) 2026

May – Aug 2026

- Contributed to **11 open-source repositories** with merged pull requests involving database migrations, security enhancements, AI engine memory optimization, bug fixes, and feature enhancements, achieving a **Global Rank of #1,206 (Top 3%)** among **43,587 participants** with a **Leaderboard Score of 3,483**.
- Collaborated with maintainers through GitHub workflows, code reviews, issue discussions, and cross-project open-source development.

Achievements & Certifications

- Solved **160+** algorithmic and data structure problems on **LeetCode**.
- **NPTEL (Elite):** Predictive Modelling with Applications – **79%** — **3 Credits**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
- Certified in Introduction to the Fundamentals of Databases.